



Correlation Between Pituitary Adenoma Surgery and Anxiety Disorder: Systematic Review and Meta-Analysis

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Key words

- Anxiety disorder
- Management
- Outcome
- Pituitary adenoma
- Systematic review

Abbreviations and Acronyms

ACTH: Adrenocorticotrophic hormone
CD: Cushing's disease
CI: Confidence interval
EQ-5D-5 L: EuroQol 5-Dimension, 5-Level instrument
GAD: Generalized anxiety disorder
GH: growth hormone
HADS: Hospital Anxiety and Depression Scale
HRT: Hormonal replacement therapy
I²: Higgins' index
NIH: National Institutes of Health
NFPA: Non-functional pituitary adenoma
PA: Pituitary adenoma
PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PRL: Prolactinoma
SAS: Zung Self-Rating Anxiety Scale
SCL 90 R: Symptom Checklist-90-Revised

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INTRODUCTION

Compared to the general population, people with pituitary adenomas (PAs)

■ **OBJECTIVE:** This study aims to evaluate the impact of surgical intervention on anxiety levels in patients with various types of pituitary adenoma (PA).

■ **METHOD:** A systematic review was conducted following PRISMA guidelines until October 2022, searching Embase, PubMed, Web of Sciences, and Scopus.

■ **RESULTS:** A total of 32 studies were included, encompassing 2,681 patients with the mean age of 53.33 ± 6.48 years (43.4% male). Among all subtypes, 664 diagnosed with Cushing's disease (25.8%), 612 with acromegaly (23.8%), 282 with prolactinoma (10.9%), and 969 with nonfunctional pituitary adenomas (37.6%). Pituitary insufficiency was the most common complication. Considering therapeutic modalities, 515 patients (29.8%) underwent endoscopic trans-sphenoidal surgery, while 222 (12.9%) underwent microscopic trans-sphenoidal surgery. The type of trans-sphenoidal surgery was not specified in 977 (56.6%) patients. A total of 17 studies including 1510 patients which mostly assessed anxiety using the Hospital Anxiety and Depression Scale (HADS) and Zung Self-Rating Anxiety Scale (SAS) were included in the meta-analysis. Preoperative evaluation using Hospital Anxiety and Depression Scale (HADS) questionnaire showed a pooled score of 8.27 (95%CI 4.54–12.01), while postoperative evaluation yielded a pooled score of 6.49 (95%CI 5.35–7.63), indicating no significant difference. Preoperative SAS assessment resulted in a pooled score of 50.43 (95%CI 37.40–63.45), with postoperative pooled score of 55.91 (95%CI 49.40–62.41), showing no significant difference.

■ **CONCLUSIONS:** Our analysis revealed no significant difference in anxiety scores pre- and postoperatively. While our findings suggest stability in anxiety levels following surgical intervention, it is imperative to recognize the limitations of the current evidence base. The observed lack of consensus may be influenced by factors such as the heterogeneous nature of the patient population, variations in the characteristics of pituitary adenomas, diverse therapeutic approaches, and potential confounding variables such as pre-existing mental health conditions and coping mechanisms. Further research is warranted to elucidate the nuanced relationship between surgical intervention for PA and anxiety outcomes, considering these complex interactions and employing rigorous methodologies to address potential sources of bias.

exhibit a discernible pattern of heightened anxiety-related personality traits or tendencies, probably related to the pituitary lesion and/or accompanying hormonal dysregulations and comorbidities.¹ The neuropsychiatric alterations, including anxiety and depression, were found to affect 30% of patients with PAs,

significantly impacting their well-being and quality of life.² Even after long-term remission, 54%–85% of Cushing's disease (CD) patients exhibit an elevated prevalence of affective changes such as anxiety, hyperactivity, emotional lability, melancholia, and apathy.^{3–5} These findings emphasize the importance of addressing

psychological aspects in the management of PAs, in addition to the physiological and clinical considerations.

Van der Klaauw et al. revealed that patients with a history of CD treatment are predisposed to more severe anxiety disorders.⁶ When compared to acromegalic patients, CD patients have a distinctive pattern of personal characteristics linked to more severe anxiety and low externalizing behavior.⁷ Loosen et al. discovered that generalized anxiety disorder (GAD) and panic disorder were four times more prevalent in relatives of CD patients than in relatives of patients with major depressive disorder.⁸ In terms of verbal recall, memory/recall, and mental agility, nonfunctional pituitary adenoma (NFPA) patients reported increased cognitive impairment,⁹ also shown to have elevated levels of anxiety.¹⁰ Mild to severe neurocognitive abnormalities in various aspects are frequently seen in patients with active untreated acromegaly.¹¹

High levels of growth hormone (GH) can disrupt neurotransmitter function, particularly involving serotonin and dopamine, which are known to play a role in mood regulation.¹² Alterations in these neurotransmitters can contribute to the development or exacerbation of anxiety symptoms.¹³ In acromegaly patients, anxiety and depression are linked to the female gender, macroadenomas, and radiotherapy history.¹⁴ Prolactin overproduction has been linked to a considerable decline in cognitive function.¹⁵ Based on a study, there was an increase in the prevalence of anxiety and a slight elevation in measures of anger in prolactinoma (PRL) patients observed.¹⁶ In this review, we aimed to evaluate the effect of surgical intervention on anxiety disorders in patients with different types of PAs.

MATERIAL AND METHODS

Literature Search

This paper was prepared using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement.^{17,18} PubMed, EMBASE, Scopus, and Web of Science were searched from database inception to October 11, 2022, using the search string available in

Supplementary data 1. Studies were exported to Rayyan, and duplicates were removed.

Study Selection

Pre-determined inclusion and exclusion criteria were defined. Based on the DSM-5, anxiety disorders include GAD, panic disorder, social anxiety disorder (social phobia), specific phobias, and agoraphobia.¹⁹ Studies were included if they involved: 1) one or more patients with one of the main five categories of anxiety; 2) diagnosed and surgically treated for PAs including adrenocorticotrophic hormone (ACTH)-secreting PA, GH-secreting PA, PRL, and NFPA; 3) regardless of anxiety severity and underwent surgical resection will be included if 4) there were written in English. Studies were excluded if they were: 1) reviews, animal studies, letters to the editor, expert opinions, operative video without any patient description, non-English records, response to the comments, conference abstracts, editorials; 2) studies involving patients with psychological problems other than anxiety such as bipolar disorder, schizophrenia, and mood disorder; 3) patients who received radio-therapeutical treatment modalities for their PA as the first-line treatment; 4) patients with any contraindication to surgical resection; and 5) Studies without data on anxiety status assessed by any type of anxiety assessment tool either before or after surgery in patients with PA were excluded from our study. However, studies with available data on anxiety status either before, after, or both before and after surgery were included in the study. In the case of studies with overlapping cohorts, only the most recent were included. Two independent reviewers (L.K. and S.S.) screened titles and abstracts of all collected articles and surveyed full texts of those meeting the inclusion criteria. Any disagreements were solved by another author's opinion (O.Y.). Eligible articles were included upon the predefined criteria, and references were searched to retrieve additional relevant studies.

Data Extraction

Data extraction was performed based on a predefined protocol by one author (L.K.) and was rechecked by another (S.S.). Disagreements were resolved by a third

author (M.S.). The extracted data for patients with PA who has anxiety disorder included various demographic information from the samples such as age, sex, type of tumor, type of surgery, and data regarding anxiety status either pre- or post-operatively. Finally, data regarding the remission after any kind of surgery, along with data regarding correspondence complications in patients with PA, was extracted and analyzed. We included only full-text articles with data on anxiety measurement in patients with PAs. Since anxiety in PAs has not been in the scope of randomized clinical trials so far, and most data comes from studies with small sample sizes, we included case reports and case series in our data analysis.

Data Synthesis

Descriptive results for categorical variables are presented as frequency (percentage), while for continuous variables, they are reported as mean and standard deviation. A meta-analysis was conducted on the quantitative anxiety scores, which were assessed using Hospital Anxiety and Depression Scale (HADS) or The Zung Self-Rating Anxiety Scale (SAS) questionnaires. To account for heterogeneity among the settings of the various studies, a random-effects model was employed. The pooling of data was performed using RStudio software version 1.2.5042 (RStudio Team, 2020) and the Meta package v4.17-0. The pooled rates are depicted in forest plots, displaying estimated proportion points with a 95% confidence interval (CI). The degree of heterogeneity between studies was assessed using Higgins' index (I^2). To evaluate potential publication bias, Egger's test and funnel plots were utilized. It is important to note that in studies reporting scores for two different gender or intervention groups, each subgroup was treated as a separate entity for the analysis.

Quality Assessment

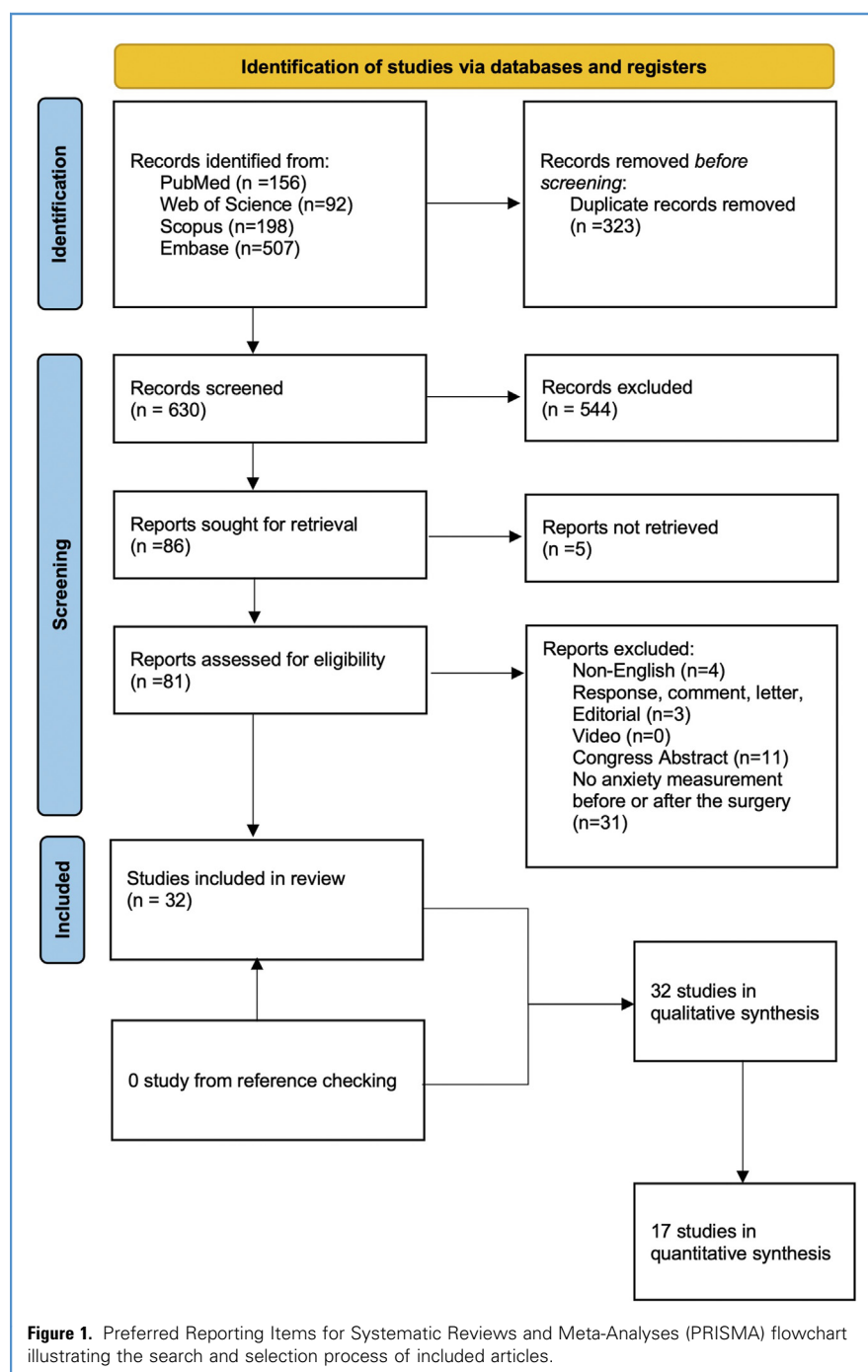
Two authors (L.K and S.S) independently assessed the quality of the included studies based on the National Institutes of Health (NIH) Quality Assessment Tool for Case Series Studies and the disagreements were resolved by a third author (M.S.) (**Supplementary data 2**).²⁰ Nine questions are posed by the NIH Quality Assessment Tool for Case Series

Studies. This scale assesses the quality of an article based on selection, comparability, and exposure giving studies a score out of nine. National Heart, Lung, and Blood Institute tool assisted in determining the “internal validity of studies” and identifying potential sources of bias.²⁰

RESULTS

Study Selection

Figure 1 displays the literature search and article selection process. A total of 32 studies were finally identified to meet the inclusion criteria, reporting anxiety assessment in patients with PAs.^{6,21-36}



Demographics and Clinicopathological Features

The systematic review encompassed 2681 patients with PAs (43.4% male) and a mean age of 53.33 ± 6.48 years. Various histology subtypes were reported in 2575 (96.0%) of these patients, while 106 were not reported (**Table 1**). Among the subtypes, NFPA was the most prevalent (37.6%), followed by GH-secreting PA (23.8%), ACTH-secreting PA (25.8%), and PRL (10.9%). Other subtypes included 48 patients (1.9%) with the diagnosis of thyrotropin-, gonadotropin-, and multi-hormonal-secreting PAs.

Therapeutic Modalities

As shown in **Table 2**, out of the patients, 515 (29.8%) underwent endoscopic trans-sphenoidal surgery, while 222 (12.9%) underwent microscopic trans-sphenoidal surgery. The type of trans-sphenoidal surgery was not specified in 977 (56.6%) patients. A mere 12 patients (0.7%) received transcranial surgery. Additionally, 213 patients (12.3%) received concomitant radiotherapy with a mean dose of 45 ± 5 Gy.

Table 1. Summary of Demographics of all Pooled Patients

Characteristics	Value
Cohort size (no.)	2681
Demographics (n = 2681)	
Mean age ($\pm$ SD)	53.33 $\pm$ 6.48 years
Gender (male)	1163 (43.4%)
Pituitary Adenoma Histology (n = 2575)	
ACTH tumour	664 (25.8%)
GH	612 (23.8%)
Prolactinoma	282 (10.9%)
Nonfunctional	969 (37.6%)
Thyrotropin	14 (0.5%)
Gonadotropin	24 (0.9%)
Multi-hormonal	10 (0.5%)
ACTH, Adrenocorticotrophic hormone-secreting pituitary adenoma; GH, Growth hormone-secreting pituitary adenoma.	

Table 2. Pituitary Adenoma Management

Therapeutic Management	
Surgical approach (n = 1726)	
Trans-sphenoidal (unspecified)	977 (56.6%)
Trans-sphenoidal (endoscopic)	515 (29.8%)
Trans-sphenoidal (microscopic)	222 (12.9%)
Transcranial	12 (0.7%)
Concomitant Radiotherapy	213 (12.3%)

Postoperative Complications

The post-operative complications of all pooled patients are concisely summarized in **Table 3**, with a total of 911 incidence. A prominent observation is the prevalence of pituitary insufficiency, with 376 cases (41.4%) categorized as unspecified. Among the specifically identified types of pituitary insufficiency, growth hormone (GH) insufficiency was observed in 164 cases (18.0%), followed by cortisol insufficiency in 121 cases (13.3%). Other complications included transient diabetes insipidus in 60 cases (6.6%), disorders of sodium metabolism in 63 cases (6.9%), and disorders of potassium metabolism in 25 cases (2.7%). Cerebrospinal fluid (CSF) leak accounted for 24 cases (2.6%),

Table 3. Summary of Postoperative Complications of all Pooled Patients

Post-therapy Complications (n = 911)	
Transient Diabetes Insipidus	60 (6.6%)
Disorders of sodium metabolism	63 (6.9%)
Disorders of potassium metabolism	25 (2.7%)
Cerebrospinal fluid rhinorrhoea	24 (2.6%)
Infection	22 (2.4%)
Pituitary insufficiency — (Unspecified)	376 (41.4%)
Pituitary insufficiency — Cortisol insufficiency	121 (13.3%)
Pituitary insufficiency — GH insufficiency	164 (18.0%)
Persistent disease	24 (2.6%)
Vision loss	32 (3.5%)
GH, Growth hormone.	

infections for 22 cases (2.4%), persistent disease for 24 cases (2.6%), and vision loss for 32 cases (3.5%).

Anxiety Assessment Tool

The summary of anxiety validation instruments is presented in **Table 4**. The total count of anxiety tools listed as 33 is because of one Psaras et al. employing both HADS and Symptom Checklist-90-Revised (SCL 90 R) assessment tools.³⁷ SAS was implemented in 7 studies (21.2%), HADS-A was utilized in 15 studies (45.5%), and the Beck Anxiety Inventory (BAI) was utilized in 1 study (3.0%). Additionally, the EuroQol 5-Dimension, 5-Level instrument (EQ-5D-5 L) was chosen in 3 studies (9.1%), while the State Trait Anxiety Inventory was employed in 2 studies (6.1%). It is noteworthy that 4 studies (12.1%) did not specify the use of any particular anxiety scale for validation.

Synthesis of the Results

Our meta-analysis included a total of 17 articles that assessed anxiety levels in patients undergoing surgical intervention for PA resection.^{6,21-36} A total of 1510 patients were included, with 750 females (49.7%). Among the 1368 cases reporting adenoma types, 297 (21.7%) had CD, 208 (15.2%) had GH imbalance, 204 (14.9%) had PRL, and 652 (47.7%) had NFPA. Surgical resection was performed in all patients, with 151 (10.2%) receiving radiotherapy^{6,23,26,28-30,37-42} as secondary adjuvant therapy, and 16 patients undergoing hormonal replacement therapy (HRT) or adrenalectomy.^{6,28-30,43} Most reports evaluated the impact of secondary interventions, such as holistic care and predefined nursery care, on alleviating anxiety-related burdens experienced by the patients.

The analysis focused on anxiety scores following the surgical intervention and prior to any secondary care programs. Different questionnaires were used for the quantitative assessment of anxiety, with the HADS and SAS being the most common, and consequently, meta-analysis performed on these 2 assessment tools. Some reports conducted the assessment either before or after the operation, depending on the cross-sectional design of the study.

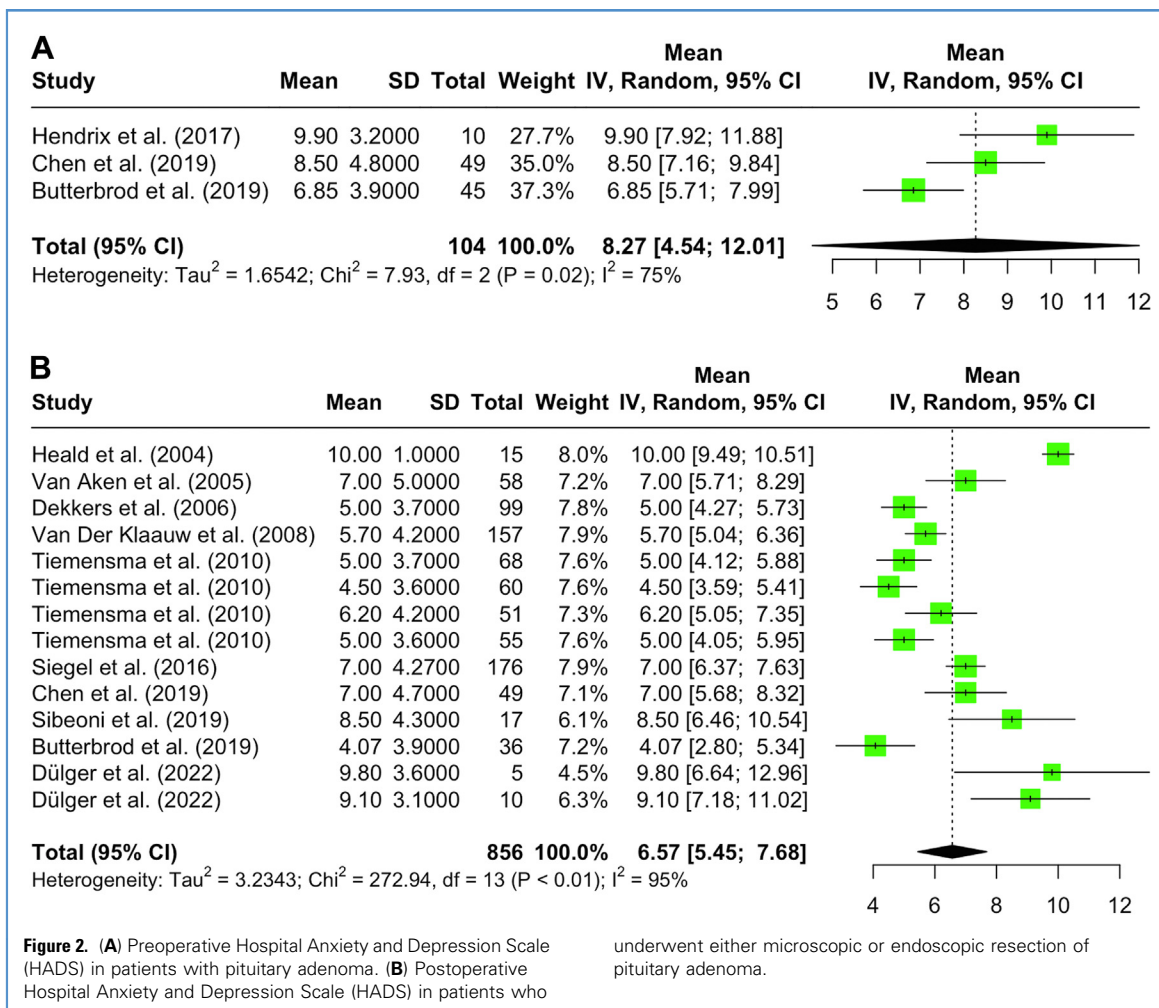
HADS. Before surgery, anxiety levels were evaluated using the HADS questionnaire in a group of 104 patients with a mean age of 49.67 years, consisting of NFPA and ACTH-producing PAs.^{21,22,27} The meta-analysis revealed a pooled HADS score of 8.27 (95%CI 4.54; 12.01) (**Figure 2A**). After surgery, a total of 856 patients were evaluated, with mean age of 53.60 years, mostly consisting of ACTH-producing PAs (47%) and NFPAs (41%).^{6,21-24,26,28-30,36,39} The pooled HADS score for patients assessed after the surgery was 6.57 (95%CI 5.45; 7.68) (**Figure 2B**). Although there was a decrease in the score after the surgery, no significant difference was reached.

SAS. Using the SAS questionnaire, anxiety levels were assessed in 349 patients with a mean age of 34.42 years before surgery, mostly consisting of NFPA (188; 53.9%), followed by GH-producing PA (67; 19.2%) and PRL (58; 16.6%).^{31,32,35} The pooled data showed a preoperation score of 50.48 (95%CI 40.64; 60.33) (**Figure 3A**), while the postoperation pooled score among 654 cases was 56.38 (95%CI 51.00; 61.76) (**Figure 3B**), with no statistically significant difference compared to the preoperation score.^{25,31-34}

Other Assessment Tools. Due to the cross-sectional design of some studies and the use of different assessment tools with insufficient data to pool assessment tools other than HADS and SAS, a comparison

Table 4. Summary of Utilized Anxiety Validation Instrument

Anxiety Instrument (n = 33)	
SAS	7 (21.2%)
HADS-A	15 (45.5%)
BAI	1 (3.0%)
EQ-5D-5 L	3 (9.1%)
State Trait Anxiety Inventory	2 (6.1%)
SCL 90 R	1 (3.0%)
No scale used	4 (12.1%)
BAI, Beck Anxiety Inventory; EQ-5D-5 L, EuroQol 5-Dimension, 5-Level instrument; HADS-A, Anxiety Subscale of the Hospital Anxiety and Depression Scale. SAS, Self-Rating Anxiety Scale; SCL 90 R, Symptom Checklist-90-Revised.	



between pre- and postoperation states could not be conducted. The unrepresentative data and the use of different questionnaires also hindered the performance of a subgroup analysis between different types of pituitary adenomas.

Quality Assessment and Publication Bias

Two authors rated the studies based on the NIH quality assessment tool. Excluding 4 case reports, the median total NIH score for the included studies was 7, with a range of 2–8. Seventeen studies were classified as good quality, with NIH scores of 7 or higher, 9 studies were classified as fair quality with NIH scores between 5 and 6, and 2 studies were classified as poor quality (NIH score ≤ 4) (Supplementary data 2). The reasons for assigning poor quality to these studies are described in the

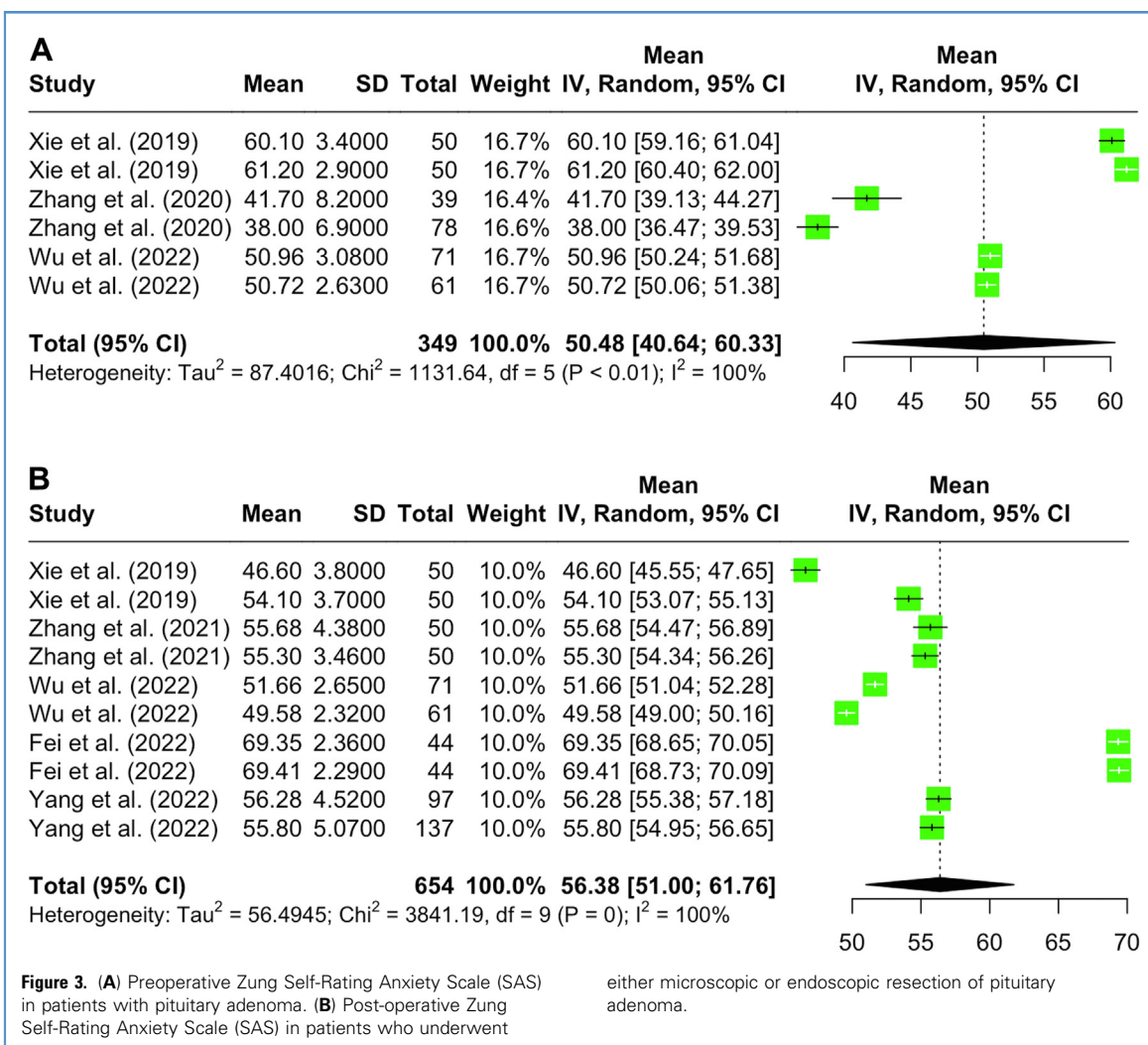
supplementary file. The authors reached a consensus on the NIH scores through discussion. Publication bias for studies which reported pre- and post-operative SAS and HADS, respectively was demonstrated on Figure 4.

DISCUSSION

In this review, we found a lack of consensus regarding the impact of surgical intervention on anxiety levels in patients with PAs. The assessment of anxiety using HADS and SAS demonstrated no statistically significant difference between preoperative and postoperative anxiety scores. However, there seems to be a discrepancy in the results obtained from these 2 assessment tools in a way that comparing the post-operative score with

the preoperative score in HADS and SAS anxiety measurement revealed decrease and increase scores, respectively. The HADS and SAS are both widely used self-report measures to assess anxiety levels in clinical populations.^{44,45} However, they differ in terms of their structure, content, and scoring methods. This variation could potentially contribute to the observed discrepancy in the results.

One possible reason for the observed discrepancy is the difference in the focus and content of the two scales. The HADS is a multidimensional scale that assesses both anxiety and depression symptoms,⁴⁶ while the SAS specifically targets anxiety symptoms.⁴⁷ As a result, the HADS may capture a broader range of emotional experiences and symptoms compared to the SAS, which focuses solely on anxiety.



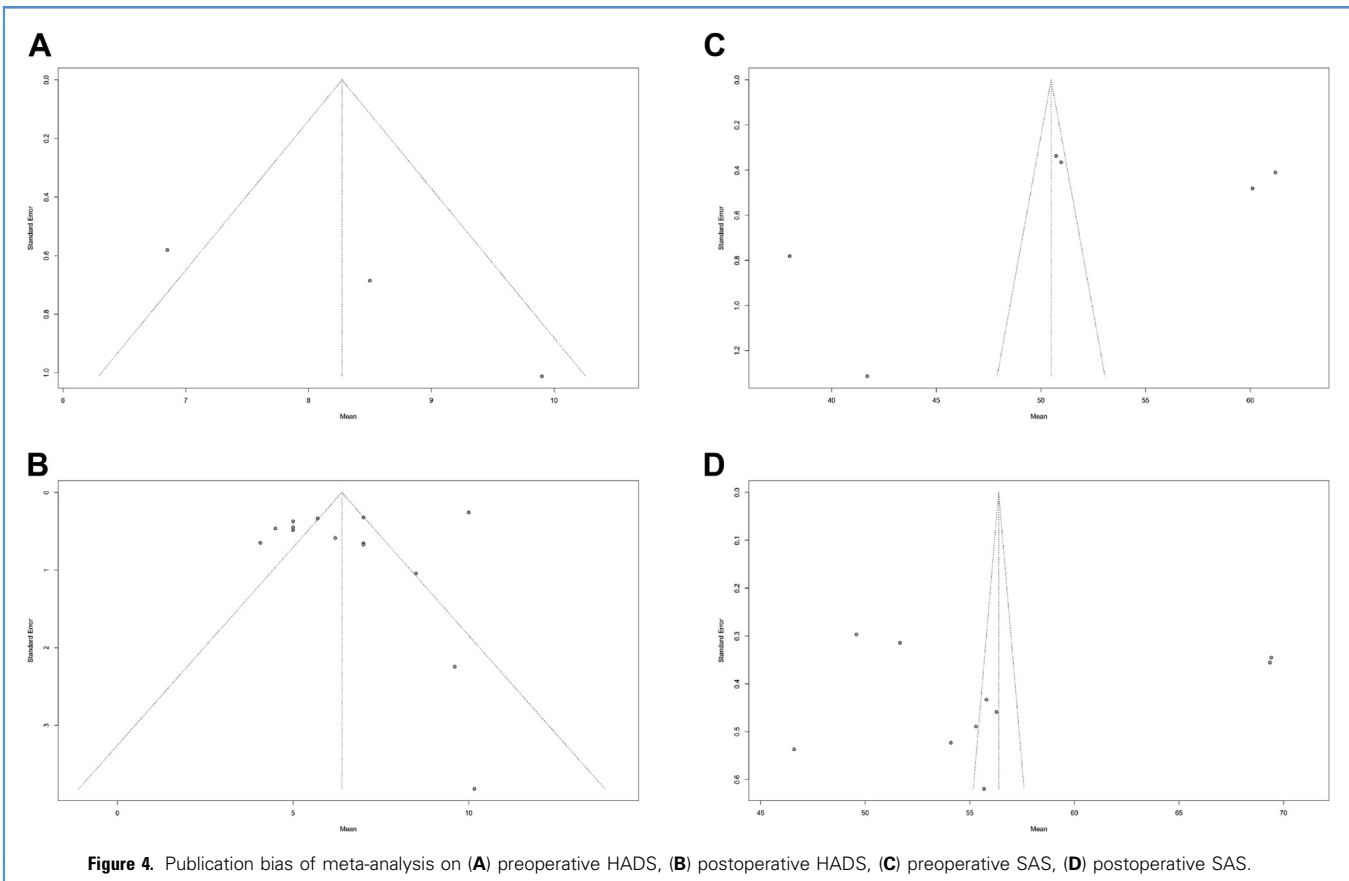
This broader focus in the HADS may dilute the specificity of anxiety symptoms and could potentially explain the slightly lower anxiety scores observed compared to the SAS. Another factor to consider is the wording and structure of the items in each scale. The HADS uses a Likert scale ranging from 0 to 3 for each item, resulting in a total score ranging from 0 to 42.⁴⁸ In contrast, the SAS employs a self-rating system with items that directly measure the intensity of anxiety symptoms.⁴⁹ The different response formats and scoring methods may influence how individuals interpret and respond to the items. Probably, the HADS's multi-item Likert scale format allows for a more nuanced evaluation of anxiety and depression symptoms, potentially leading

to slightly different results compared to the SAS. Additionally, the composition of the study sample and the specific characteristics of the individuals included may contribute to the observed discrepancy.

The heterogeneity of the population, including variations in age, gender, PA types, tumor size, and symptom duration, could potentially influence the way individuals experience and report anxiety symptoms. Although the size of tumor and duration of symptoms in most patients were not reported, larger tumors and longer durations of symptoms supposed to increased anxiety due to their impact on neurologic and endocrine function.⁵⁰ For instance, the presence of visual disturbances, headaches, or hormonal imbalances associated with

larger adenomas could contribute to heightened anxiety levels. Other factors that were not described in included studies were substance use and mental health history. Unfortunately, substance use issues are becoming increasingly common in society, and they can interact with anxiety in complex ways.⁵¹ Additionally, a patient's personal and family mental health history may play a role in their susceptibility to anxiety, requiring careful consideration in future research. It is essential to consider these factors when interpreting the results obtained from different assessment tools.

Among the reported cases, NFPAs, CD, GH-secreting PA, and PRL were the most common types. The heterogeneity in PA types may contribute to variations in



anxiety levels and outcomes. It might be difficult to detect NFPA's since they do not exhibit hormonal hypersecretion. Numerous testing and imaging examinations are frequently required for the diagnosis of NFPA's, which causes patients to experience a time of uncertainty and worry.⁵² Contrarily, more precise hormonal irregularities help in the identification of conditions including acromegaly, PRL, and CD, possibly lowering diagnostic ambiguity and related anxiety. Furthermore, visual problems, headaches, and other neurologic symptoms may result from the NFPA's size and location, and these symptoms may increase anxiety.⁵³ Concerns about the development and management of the disease may further be exacerbated by the chronic nature of NFPA's and the possible requirement for long-term surveillance.

Based on our review, among hormone-producing PAs, CD exhibited the highest

prevalence. Overproduction of cortisol, a stress hormone that can affect mood and anxiety levels, is a hallmark of CD.⁷ Increased anxiety, cognitive impairment and psychological distress might result from high cortisol levels that interfere with the stress response system's ability to be regulated.⁵⁴ Acromegaly and PRL could not have the same immediate physiological effects on regulating anxiety as CD. Body image, self-esteem, and general well-being can be greatly impacted by CD symptoms such as weight gain, central obesity, moon face, buffalo hump, and skin abnormalities.⁵⁵⁻⁵⁷ These obvious physical alterations might raise anxiety levels in CD patients, as can other symptoms including exhaustion, muscular weakness, and mood fluctuations.^{58,59} While PRL and acromegaly may have their own physical symptoms, they might not be as obvious or upsetting psychologically as CD. Further research is needed to explore whether different PA

types are associated with distinct patterns of anxiety and if specific factors influence anxiety levels in patients with different PA subtypes.

The surgical approach varied across the included studies, with 29.8% of cases undergoing endoscopic surgical resection and 12.9% undergoing microscopic surgical resection and the surgical approach was not specified in 56.6% of patients. The impact of different surgical techniques on anxiety levels remains unclear and warrants further investigation. It would be valuable to explore whether the occurrence of these complications contributes to anxiety in patients with PAs.

The use of adjuvant radiotherapy and/or HRT was reported in a subset of cases. Radiotherapy can have a variety of effects on anxiety levels. Radiation may provide patients a feeling of security by concentrating on any cancer cells still present and lowering the possibility of recurrence.⁶⁰ This could reduce stress brought on by

the condition. However, the possible side effects of radiation, such as fatigue, skin changes, and long-term risks, might also raise concerns and anxiety.⁶¹ Overcoming these fears can be facilitated by psychological support along with effective communications with healthcare providers. In addition, appropriate HRT might assist in restoring hormonal balance if the adenoma or its treatment caused abnormalities in cortisol, GH, thyroid hormone, or sex hormones. This return to hormonal equilibrium may benefit mood, energy, and general well-being while perhaps alleviating anxiety symptoms.⁶²

The findings of this review highlight the need for more comprehensive research on the relationship between anxiety and PAs. Future studies should aim to identify potential risk factors for anxiety development in patients with PAs, such as pre-existing psychological conditions, tumor characteristics, and treatment-related factors. Longitudinal studies assessing anxiety levels at multiple time points and incorporating validated psychological assessment tools will provide a more nuanced understanding of the temporal changes and long-term impact of surgical intervention on anxiety in these patients.

Improved understanding of the relationship between anxiety and PAs will facilitate the development of targeted interventions and support strategies. Collaborative efforts between neurosurgeons, endocrinologists, psychiatrists, and psychologists are crucial in addressing the multidimensional aspects of patient care. A multidisciplinary approach that integrates psychological support and interventions alongside medical and surgical management may be necessary to optimize the well-being and quality of life of individuals with PAs and associated anxiety.

Strength and Limitation

To the best of our knowledge, this study represents the first comprehensive review focusing on the association between PA and anxiety disorder, providing quantitative results. However, it is important to acknowledge certain limitations. Our investigation included a variety of articles with different study designs, sample sizes, methodologies, and levels of quality. The variety of the studies that were included

restricts the generalizability of our findings and introduces possible sources of bias. Direct comparison and combination of the data are challenging due to the differences in research characteristics, which might affect the overall validity of our findings. The use of different anxiety assessment instruments including HADS and SAS increases the risk of measuring variability and consistency when attempting to accurately quantify the level of anxiety present in individuals with PA. Although we covered a variety of PAs, our study ran into issues when trying to perform a thorough subgroup analysis. Our ability to investigate and contrast anxiety levels among different PA types was precluded by unrepresentative data and the use of heterogeneous questionnaires. This constraint prevents a thorough assessment of the distinctive effects of various PA types on anxiety outcomes.

We acknowledge the limitations stemming from the high rates of heterogeneity observed across various subsections of our meta-analysis, necessitating caution in interpreting the results. Due to the absence of reported data in the included studies, we were unable to conduct sensitivity analysis or subgroup analysis to explore the impact of potential confounding factors, including previous mental health problems, quality of life, surgical complications, etc., on anxiety levels in patients undergoing PA surgery. As a result, our study primarily focused on measuring anxiety levels before and after surgery, rather than thoroughly assessing other factors that might contribute to anxiety in PA patients, such as previous mental health problems, coping mechanisms, social support, and quality of life.

CONCLUSION

In conclusion, our study contributes to the growing body of literature examining the relationship between surgical intervention for pituitary adenomas and anxiety levels. While our analysis did not find a significant difference in anxiety scores pre- and postoperatively based on HADS and SAS assessments, it is essential to interpret these findings in context. The observed lack of consensus may be attributed to various factors, including the heterogeneous nature of the patient population, variability in tumor characteristics, diverse

therapeutic modalities, and potential confounding variables such as pre-existing mental health conditions and coping mechanisms. These factors can influence anxiety outcomes and contribute to discrepancies between studies. Moving forward, future research should employ longitudinal study designs, incorporate validated psychological assessment tools, and consider these multifaceted factors to provide a more comprehensive understanding of the complex interplay between surgical intervention and anxiety in patients with pituitary adenomas. Collaborative efforts between clinicians, researchers, and mental health professionals are essential to optimize patient care and support strategies in this population.

CRediT AUTHORSHIP CONTRIBUTION STATEMENT

Mohammadmahdi Sabahi: Data curation, Formal analysis, Methodology, Writing – original draft. **Omid Yousefi:** Investigation, Methodology, Resources, Writing – review & editing. **Laura Kehoe:** Data curation, Formal analysis, Resources, Writing – review & editing. **Sarvin Sasannia:** Data curation, Formal analysis, Resources, Writing – review & editing. **Clayton Gerndt:** Resources, Visualization, Writing – review & editing. **Badih Adada:** Resources, Writing – review & editing. **Hamid Borghei-Razavi:** Resources, Writing – review & editing. **Kiarash Shahlaie:** Resources, Visualization, Writing – review & editing. **Paolo Palmisciano:** Writing – review & editing, Conceptualization, Project administration, Resources, Supervision, Visualization.

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SUPPLEMENTARY MATERIAL

Supplementary Table 1. Search Strategy

Database	Search Strategy
PubMed	"Anxiety"[Mesh] OR "Anxiety, Separation"[Mesh] OR "Anxiety Disorders"[Mesh] OR "Phobia, Social"[Mesh] OR "Phobic Disorders"[Mesh] OR "Panic"[Mesh] OR "Panic Disorder"[Mesh] OR "Anxiety"[Title/Abstract] OR "Separation anxiety"[Title/Abstract] OR "Anxiety Disorder"[Title/Abstract] OR "Social Phobia"[Title/Abstract] OR "Phobia"[Title/Abstract] OR "Phobic disorder"[Title/Abstract] OR "Panic"[Title/Abstract] OR "Panic disorder"[Title/Abstract]
	AND "Surgical Procedures, Operative"[Mesh] OR "Neurosurgical Procedures"[Mesh] OR "operation"[Title/Abstract] OR "procedure"[Title/Abstract] OR "surgery"[Title/Abstract] OR "surgical"[Title/Abstract] OR "neurosurgery"[Title/Abstract] OR "resection"[Title/Abstract] OR "excision"[Title/Abstract]
	AND "Pituitary Neoplasms"[Mesh] OR "ACTH-Secreting Pituitary Adenoma"[Mesh] OR "Growth Hormone-Secreting Pituitary Adenoma"[Mesh] OR "Prolactinoma"[Mesh] OR "Pituitary Adenoma"[Mesh] OR "Pituitary Neoplasms"[Title/Abstract] OR "ACTH-Secreting Pituitary Adenoma"[Title/Abstract] OR "ACTH Secreting Pituitary Adenoma"[Title/Abstract] OR "Adrenocorticotrophic hormone-Secreting Pituitary Adenoma"[Title/Abstract] OR "Adrenocorticotrophic hormone Secreting Pituitary Adenoma"[Title/Abstract] OR "Growth Hormone-Secreting Pituitary Adenoma"[Title/Abstract] OR "Growth Hormone Secreting Pituitary Adenoma"[Title/Abstract] OR "Prolactinoma"[Title/Abstract] OR "Prolactin-Secreting Pituitary Adenoma"[Title/Abstract] OR "Prolactin Secreting Pituitary Adenoma"[Title/Abstract] OR "Pituitary Adenoma"[Title/Abstract] OR "Cushing disease"[Title/Abstract] OR "Cushing's disease"[Title/Abstract] OR "Acromegaly"[Title/Abstract] OR "Pituitary-Dependent Hypercortisolism"[Title/Abstract] OR "Pituitary Dependent Hypercortisolism"[Title/Abstract] OR "Pituitary Dependent Cushing Syndrome"[Title/Abstract] OR "Pituitary-Dependent Cushing Syndrome"[Title/Abstract] OR "corticotrophic pituitary adenoma"[Title/Abstract] OR "GH secreting pituitary adenoma"[Title/Abstract] OR "GH-secreting pituitary adenoma"[Title/Abstract] OR "Non-functional pituitary adenoma"[Title/Abstract] OR "NFPA"[Title/Abstract] OR "ACTH-producing pituitary adenoma"[Title/Abstract] OR "ACTH producing pituitary adenoma"[Title/Abstract] OR "Adrenocorticotrophic hormone-producing pituitary adenoma"[Title/Abstract] OR "Adrenocorticotrophic hormone producing pituitary adenoma"[Title/Abstract] OR "GH-producing pituitary adenoma"[Title/Abstract] OR "GH producing pituitary adenoma"[Title/Abstract] OR "Growth hormone-producing pituitary adenoma"[Title/Abstract] OR "Growth hormone producing pituitary adenoma"[Title/Abstract] OR "prolactin producing pituitary adenoma" OR "prolactin-producing pituitary adenoma"[Title/Abstract] OR "Hypophysis Neoplasms"[Title/Abstract] OR "Hypophysis Adenoma"[Title/Abstract] OR "Hypophysis tumor"[Title/Abstract]
Scopus	AND TITLE-ABS-KEY ("Anxiety") OR TITLE-ABS-KEY ("Separation anxiety") OR TITLE-ABS-KEY ("Anxiety Disorder") OR TITLE-ABS-KEY ("Social Phobia") OR TITLE-ABS-KEY ("Phobia") OR TITLE-ABS-KEY ("Phobic disorder") OR TITLE-ABS-KEY ("Panic") OR TITLE-ABS-KEY ("Panic disorder")
	TITLE-ABS-KEY ("surg*") OR TITLE-ABS-KEY ("neurosurg*") OR TITLE-ABS-KEY ("operat*") OR TITLE-ABS-KEY ("procedure*") OR TITLE-ABS-KEY ("surgical") OR TITLE-ABS-KEY ("resection") OR TITLE-ABS-KEY ("excision")
	AND TITLE-ABS-KEY ("Pituitary Neoplasms") OR TITLE-ABS-KEY ("ACTH-Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("ACTH Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Adrenocorticotrophic hormone-Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Adrenocorticotrophic hormone Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Growth Hormone-Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Growth Hormone Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Prolactinoma") OR TITLE-ABS-KEY ("Prolactin-Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Prolactin Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Pituitary Adenoma") OR TITLE-ABS-KEY ("Pituitary Neoplasms") OR TITLE-ABS-KEY ("Cushing disease") OR TITLE-ABS-KEY ("Cushing's disease") OR TITLE-ABS-KEY ("Acromegaly") OR TITLE-ABS-KEY ("Pituitary-Dependent Hypercortisolism") OR TITLE-ABS-KEY ("ACTH-Secreting Pituitary Adenoma") OR TITLE-ABS-KEY ("Pituitary Dependent Hypercortisolism") OR TITLE-ABS-KEY ("Pituitary Dependent Cushing Syndrome") OR TITLE-ABS-KEY ("Pituitary-Dependent Cushing Syndrome") OR TITLE-ABS-KEY ("corticotrophic pituitary adenoma") OR TITLE-ABS-KEY ("GH secreting pituitary adenoma") OR TITLE-ABS-KEY ("GH-secreting pituitary adenoma") OR TITLE-ABS-KEY ("Nonfunctional pituitary adenoma") OR TITLE-ABS-KEY ("NFPA") OR TITLE-ABS-KEY ("ACTH-producing pituitary adenoma") OR TITLE-ABS-KEY ("ACTH producing pituitary adenoma") OR TITLE-ABS-KEY ("Adrenocorticotrophic hormone-producing pituitary adenoma") OR TITLE-ABS-KEY ("Adrenocorticotrophic hormone producing pituitary adenoma") OR TITLE-ABS-KEY ("GH-producing pituitary adenoma") OR TITLE-ABS-KEY ("GH producing pituitary adenoma") OR TITLE-ABS-KEY ("Growth hormone-producing pituitary adenoma") OR TITLE-ABS-KEY ("Growth hormone producing pituitary adenoma") OR TITLE-ABS-KEY ("prolactin producing pituitary adenoma") OR TITLE-ABS-KEY ("prolactin-producing pituitary adenoma") OR TITLE-ABS-KEY ("Hypophysis Neoplasms") OR TITLE-ABS-KEY ("Hypophysis Tumor") OR TITLE-ABS-KEY ("Hypophysis Adenoma")
Continues	

Supplementary Table 1. Continued

Database	Search Strategy
Web of Science	AND TS=(“Anxiety” OR “Separation anxiety” OR “Anxiety Disorder” OR “Social Phobia” OR “Phobia” OR “Phobic disorder” OR “Panic” OR “Panic disorder”)
	TS=(“surg*” OR “neurosurg*” OR “operat*” OR “procedure*” OR “surgical” OR “resection” OR “excision”)
	AND TS=(“Pituitary Neoplasms” OR “ACTH-Secreting Pituitary Adenoma” OR “ACTH Secreting Pituitary Adenoma” OR “Adrenocorticotrophic hormone-Secreting Pituitary Adenoma” OR “Adrenocorticotrophic hormone Secreting Pituitary Adenoma” OR “Growth Hormone-Secreting Pituitary Adenoma” OR “Growth Hormone Secreting Pituitary Adenoma” OR “Prolactinoma” OR “Prolactin-Secreting Pituitary Adenoma” OR “Prolactin Secreting Pituitary Adenoma” OR “Pituitary Adenoma” OR “Cushing disease” OR “Cushing’s disease” OR “Acromegaly” OR “Pituitary-Dependent Hypercortisolism” OR “ACTH-Secreting Pituitary Adenoma” OR “Pituitary Dependent Hypercortisolism” OR “Pituitary Dependent Cushing Syndrome” OR “Pituitary-Dependent Cushing Syndrome” OR “corticotrophic pituitary adenoma” OR “GH secreting pituitary adenoma” OR “GH-secreting pituitary adenoma” OR “Nonfunctional pituitary adenoma” OR “NFPA” OR “ACTH-producing pituitary adenoma” OR “ACTH producing pituitary adenoma” OR “Adrenocorticotrophic hormone-producing pituitary adenoma” OR “Adrenocorticotrophic hormone producing pituitary adenoma” OR “GH-producing pituitary adenoma” OR “GH producing pituitary adenoma” OR “Growth hormone-producing pituitary adenoma” OR “Growth hormone producing pituitary adenoma” OR “prolactin producing pituitary adenoma” OR “prolactin-producing pituitary adenoma” OR “Hypophysis Neoplasms” OR “Hypophysis Adenoma” OR “Hypophysis tumor”)
EMBASE	AND “anxiety”/exp OR “anxiety disorder”/exp OR “Separation anxiety”:ti,ab,kw OR “phobia”:ti,ab,kw OR “social phobia”:ti,ab,kw OR “Phobic disorder”:ti,ab,kw OR “panic”:ti,ab,kw OR “Panic disorder”:ti,ab,kw
	“surgery”/exp OR “neurosurgery”/exp OR “surg*”:ti,ab,kw OR “operat*”:ti,ab,kw OR “resect*”:ti,ab,kw OR “excis*”:ti,ab,kw OR “procedure”:ti,ab,kw OR “surgical”:ti,ab,kw OR “neurosurg*”:ti,ab,kw
	AND “hypophysis tumor”/exp OR “Pituitary Neoplasms”:ti,ab,kw OR “ACTH-Secreting Pituitary Adenoma”:ti,ab,kw OR “ACTH Secreting Pituitary Adenoma”:ti,ab,kw OR “Adrenocorticotrophic hormone-Secreting Pituitary Adenoma”:ti,ab,kw OR “Adrenocorticotrophic hormone Secreting Pituitary Adenoma”:ti,ab,kw OR “Growth Hormone-Secreting Pituitary Adenoma”:ti,ab,kw OR “Growth Hormone Secreting Pituitary Adenoma”:ti,ab,kw OR “Prolactinoma”:ti,ab,kw OR “Prolactin-Secreting Pituitary Adenoma”:ti,ab,kw OR “Prolactin Secreting Pituitary Adenoma”:ti,ab,kw OR “Pituitary Adenoma”:ti,ab,kw OR “Cushing disease”:ti,ab,kw OR “Cushing’s disease”:ti,ab,kw OR “Acromegaly”:ti,ab,kw OR “Pituitary-Dependent Hypercortisolism”:ti,ab,kw OR “ACTH-Secreting Pituitary Adenoma”:ti,ab,kw OR “Pituitary Dependent Hypercortisolism”:ti,ab,kw OR “Pituitary Dependent Cushing Syndrome”:ti,ab,kw OR “Pituitary-Dependent Cushing Syndrome”:ti,ab,kw OR “corticotrophic pituitary adenoma”:ti,ab,kw OR “GH secreting pituitary adenoma”:ti,ab,kw OR “GH-secreting pituitary adenoma”:ti,ab,kw OR “Nonfunctional pituitary adenoma”:ti,ab,kw OR “NFPA”:ti,ab,kw OR “ACTH-producing pituitary adenoma”:ti,ab,kw OR “ACTH producing pituitary adenoma”:ti,ab,kw OR “Adrenocorticotrophic hormone-producing pituitary adenoma”:ti,ab,kw OR “Adrenocorticotrophic hormone producing pituitary adenoma”:ti,ab,kw OR “GH-producing pituitary adenoma”:ti,ab,kw OR “GH producing pituitary adenoma”:ti,ab,kw OR “Growth hormone-producing pituitary adenoma”:ti,ab,kw OR “Growth hormone producing pituitary adenoma”:ti,ab,kw OR “prolactin producing pituitary adenoma”:ti,ab,kw OR “prolactin-producing pituitary adenoma”:ti,ab,kw OR “Hypophysis Neoplasms”:ti,ab,kw OR “Hypophysis Adenoma”:ti,ab,kw OR “Hypophysis tumor”:ti,ab,kw)

Supplementary Table 2. Quality Ratings of Included Studies According to NIH Quality Assessment Tool for Case Series/case Reports Studies

Study	Was the Study Question or Objective Clearly Stated?	Was the Study Population Clearly and Fully Described, including a Case Definition?	Were the Cases Consecutive?	Were the Subjects Comparable?	Was the Intervention Clearly Described?	Were the Outcome Measures Clearly Defined, Valid, Reliable, and Implemented Consistently Across all Study Participants?	Was the Length of Follow-Up Adequate?	Were the Statistical Methods Well-Described?	Were the Results Well-Described?	Quality Rating	
1 Wu et al., 2022 ³¹	YES	YES	YES	YES	YES	NO	NO	NO	YES	7	Good
2 Fei et al., 2022 ²⁵	YES	YES	YES	YES	YES	YES	NO	YES	YES	8	Good
3 Zhang et al., 2021 ³⁴	YES	YES	NO	YES	YES	YES	NO	YES	YES	7	Good
4 Guo et al., 2021 ³⁸	YES	YES	NO	NO	YES	YES	YES	YES	NO	6	Fair
5 Zhang et al., 2020 ³⁵	YES	YES	YES	YES	YES	YES	NO	YES	YES	8	Good
6 Tan et al., 2020 ⁵¹	N/A (Case Report)										CD
7 Zhang et al., 2020 ³⁵	YES	YES	YES	NO	YES	YES	NO	YES	YES	7	Good
8 Chen et al., 2019 ²²	YES	YES	YES	YES	YES	YES	NO	YES	YES	8	Good
9 Sibeoni et al., 2019 ²⁸	YES	YES	NO	NO	NO	NO	NO	YES	NO	3	Poor
10 Daya et al., 2017 ⁴³	N/A (Case Report)										CD
11 Siegel et al., 2016 ³⁹	YES	YES	NO	NO	NO	YES	NO	YES	YES	5	Fair
12 Carluccio et al., 2015 ⁴⁰	YES	YES	YES	NO	YES	YES	NO	YES	YES	7	Good
13 Andela et al., 2015 ^{E2}	YES	NO	NO	N/A	NO	NO	NO	NO	YES	2	Poor

14	Wallace et al., 2015 ^{E3}	N/A (Case Report)										CD
15	Petry et al., 2011 ⁴¹	YES	YES	NO	YES	YES	YES	NO	YES	YES	7	Good
16	Psaras et al., 2011 ³⁷	YES	YES	NO	YES	YES	YES	YES	YES	YES	8	Good
17	Tiemensma et al., 2010 ²⁹	YES	YES	NO	YES	YES	YES	NO	YES	YES	7	Good
18	Tiemensma et al., 2010 ³⁶	YES	YES	YES	YES	YES	NO	NO	NO	YES	6	Fair
19	Van Der Klaauw et al. 2008 ⁶	YES	YES	NO	NO	YES	NO	NO	YES	YES	5	Fair
20	Van Der Klaauw et al., 2007 ^{E4}	YES	YES	YES	NO	NO	NO	YES	NO	YES	5	Fair
21	Dekkers et al., 2006 ²³	YES	YES	YES	YES	YES	YES	NO	NO	YES	7	Good
22	Van Aken et al., 2005 ³⁰	YES	YES	YES	YES	YES	NO	NO	NO	YES	6	Fair
23	Heald et al., 2004 ²⁶	YES	YES	YES	YES	YES	NO	NO	NO	YES	6	Fair
24	Peace et al., 1997 ⁴²	YES	YES	NO	YES	YES	YES	NO	YES	YES	7	Good
25	Wilcox et al., 1991 ^{E5}	N/A (Case Report)										CD
26	Xie et al., 2019 ³²	YES	YES	NO	YES	YES	YES	NO	NO	YES	6	Fair
27	Hendrix et al., 2017 ²⁷	YES	YES	YES	YES	YES	YES	NO	NO	YES	7	Good
28	Yang et al., 2022 ³³	YES	YES	YES	YES	YES	YES	NO	YES	YES	8	Good
Continues												

Supplementary Table 2. Continued

	Study	Was the Study Question or Objective Clearly Stated?	Was the Study Population Clearly and Fully Described, including a Case Definition?	Were the Cases Consecutive?	Were the Subjects Comparable?	Was the Intervention Clearly Described?	Were the Outcome Measures Clearly Defined, Valid, Reliable, and Implemented Consistently Across all Study Participants?	Was the Length of Follow-Up Adequate?	Were the Statistical Methods Well-Described?	Were the Results Well-Described?	Quality Rating	
29	Zhang et al., 2020 ³⁵	YES	NO	YES	YES	YES	YES	NO	YES	NO	6	Fair
30	Dülger et al., 2022 ²⁴	YES	YES	YES	YES	YES	YES	YES	YES	NO	8	Good
31	Butterbrod et al., 2019 ²¹	YES	YES	YES	YES	YES	YES	YES	NO	YES	8	Good
32	Wagner et al., 2019 ^{E6}	YES	YES	YES	YES	YES	YES	NO	YES	YES	8	Good

Quality Rating, ≥ 7 = Good, $5-6$ = Fair, ≤ 4 = Poor.

Two studies are deemed of poor quality: The lack of key details in the methodology and reporting in one of the studies makes it difficult to evaluate the scientific rigor and validity of the study findings (1). The other study does not clearly present robust quantitative analysis of results necessary to assess the overall evidence in addressing a specific research question (2). Consequently, the overall quality of both studies is considered poor, indicating a high risk of bias.

N/A, Not Applicable; CD, Cannot Determine.

SUPPLEMENTARY REFERENCES

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